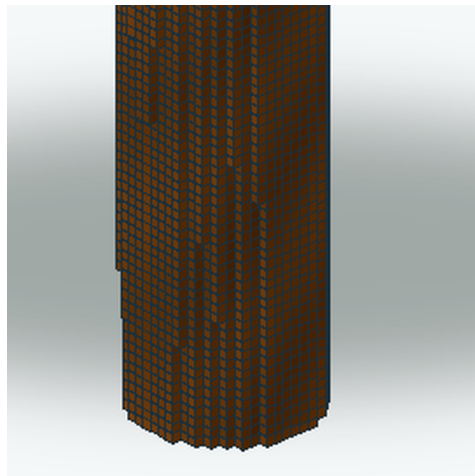
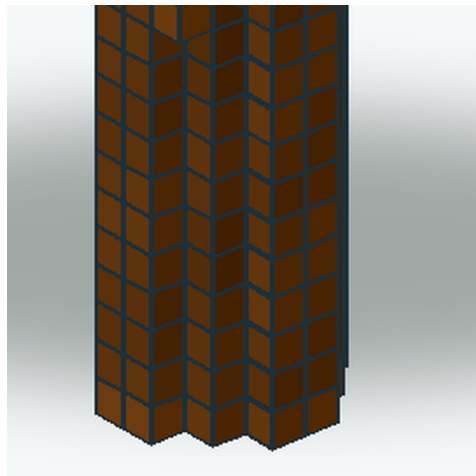


(a) Fine 128x128x128 lower trunk



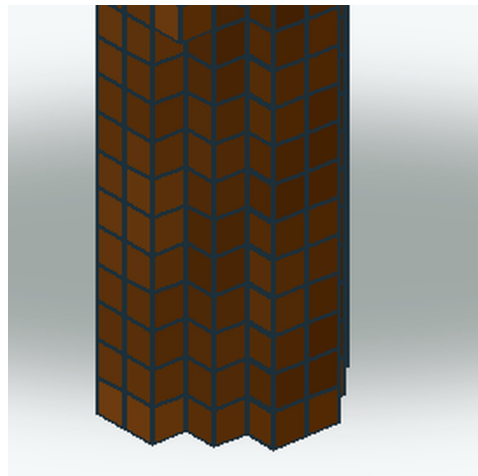
Full leafy tree retained; bottom 30% enlarged

(b) 4x4x4 lower-trunk LoD



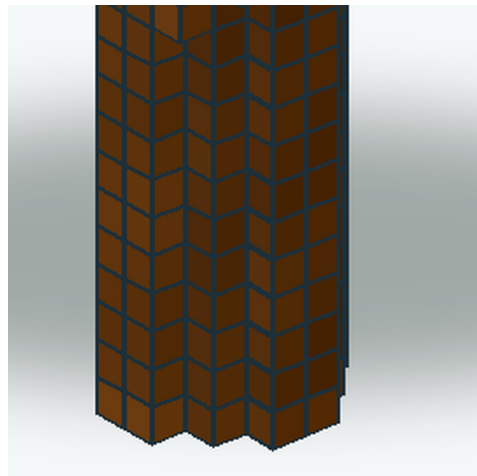
Same LoD level; box normals only

(c) Node-local relief at same LoD



Per-node quantised descriptors

(d) Relief + GGX variance



Same descriptors; modulated roughness